



Next Generation of Policy Frameworks and Modeling

WEEK 1 • JUNE 8–12, 2026

DynareJulia and FPAS Mark II:

History and Next Generation of Central Bank
Analytical Frameworks and Modeling Building

Chulalongkorn University • Bangkok, Thailand



Welcome

The Better Policy Project (BPP)

A Portugal-based institution dedicated to excellence in policymaking and economic analysis. BPP helps policy-making institutions build better and more efficient policy frameworks, improve transparency and communication, and strengthen institutional capacity.

AI-Powered Learning

Developing safe, secure, and responsible productivity tools for economists, analysts, and communication teams in central banks, ministries of finance, and financial institutions — with human oversight at the core.

8+

Years of operations

30+

Institutions

400+

Students



Week 1 at a Glance

June 8–12, 2026 · A progressive 5-day journey from institutional principles to a working model

DAY 1

Mon · Jun 8

Foundations

Why FPAS Mark II?
History, motivation,
and the folly of
baselines.

DAY 2

Tue · Jun 9

Framework

Monetary
transmission,
analytical structure,
scenario thinking.

DAY 3

Wed · Jun 10

QPMs

Semi-structural
Quarterly Projection
Models and
unobserved
variables.

DAY 4

Thu · Jun 11

DynareJulia

Implementing
QPMs; nonlinearities
and endogenous
credibility.

DAY 5

Fri · Jun 12

Capstone

Calibrate a working
model ready to
produce FPAS Mark
II scenarios.



DAY 1 • MONDAY, JUNE 8

Why FPAS Mark II?

The folly of baselines — and a paradigm shift toward risk management

THE PROBLEM

The Folly of Baselines

- False sense of assurance in a single "most-likely" path
- Centers one interpretation of a highly uncertain future
- Limits policy agility when shocks hit
- Linear models miss dangerous non-linearities
- Risk assessments and uncertainty get downplayed

THE RESPONSE

Monetary Policy as Risk Management

- Prepare for several plausible futures, not a single forecast
- Evaluate policy across scenarios to minimize regret
- Avoid "dark corners" through active risk management
- Open, structured debate replaces consensus-chasing
- Communicate uncertainty and trade-offs clearly



DAY 1 • FOUNDATIONS

Winds of Change: Rethinking Monetary Policy

Leading central bankers and economists now frame policy as risk management

Alan Greenspan

Fed, 2003

“Monetary policy in the US has come to involve, at its core, crucial elements of risk management.”

Jerome Powell

Fed, 2024

“We're really in a risk management mode — managing the risk that we move too soon and we move too late.”

Isabel Schnabel

ECB, 2024

“Systematically include alternative scenarios to communicate the uncertainty around central bank projections more directly.”

Ben Bernanke

Brookings, 2024

“Under risk management, policymakers might choose to take out some 'insurance' against bad outcomes.”

Charles Goodhart

LSE, 2023

“Central banks should present scenarios of what might happen if conditions turn out better or worse — not single point forecasts.”

Larry Summers

Harvard, 2024

“The systematic use of multiple scenarios and the desire to formulate policies that are robust across them seems to me very important.”

Three ingredients of FPAS



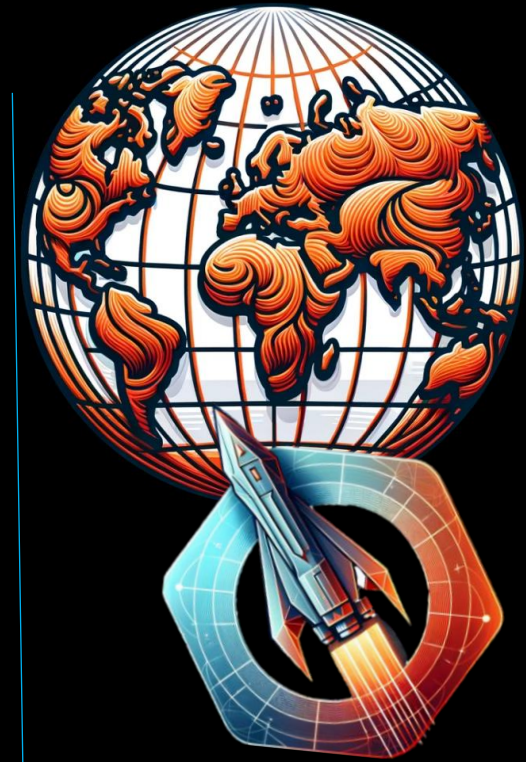
Where is the economy now?



What are the underlying forces?



What do policy instruments (monetary, fiscal) need to do to achieve inflation and other objectives?



	No FPAS	FPAS Mark I	FPAS Mark II
Three ingredients of FPAS	✗	✓	✓
Uncertainty: Scenario Analysis	✗	✗	✓
“Least Regrets” Policy	✗	✗	✓
Accountability & Transparency	✗	✓	✓



Three ingredients of FPAS



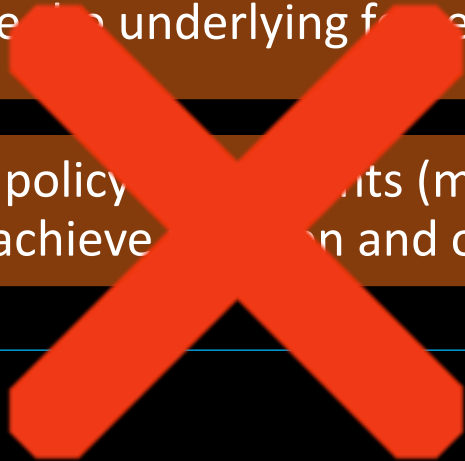
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No FPAS

Three ingredients of FPAS

Uncertainty: Scenario Analysis

“Least Regrets” Policy

Accountability & Transparency



Experiences Before FPAS, IT Lite



Very Low Levels of CB
Efficiency, Transparency
and Accountability

Three ingredients of FPAS



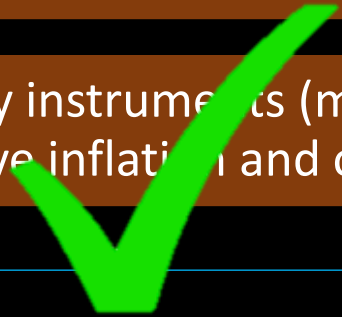
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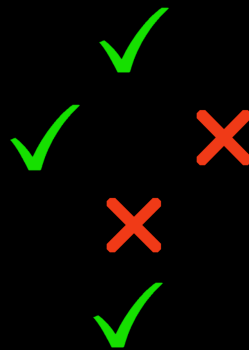
FPAS Mark I

Three ingredients of FPAS

Uncertainty: Scenario Analysis

“Least Regrets” Policy

Accountability & Transparency



Experiences with FPAS Mark I Full-Fledged Inflation Targeting



Improved CB Efficiency,
Transparency and
Accountability

Three ingredients of FPAS



Where is the economy now?



What are the underlying forces?



What do policy instruments (monetary, fiscal) need to do to achieve inflation and other objectives?

New FPAS Mark II



FPAS Mark II

Three ingredients of FPAS



Uncertainty: Scenario Analysis



“Least Regrets” Policy



Accountability & Transparency



More Improvements in CB
Efficiency, Transparency
and Accountability



DAY 2 • TUESDAY, JUNE 9

Monetary Transmission Mechanism

From shocks to objectives — the analytical map behind policy decisions

SHOCKS

Hitting the economy

- Financial shocks (foreign rates, portfolio shifts)
- Aggregate demand (credit, foreign demand, commodities, fiscal)
- Potential output shocks
- Inflation shocks (indirect taxes, energy prices)

INSTRUMENTS

Policy levers

- Policy rate path
- Longer-term interest rates
- Exchange rate channel
- Unconventional tools (QE, ELB, fiscal coordination)

OBJECTIVES

Stabilization targets

- Inflation
- Aggregate demand
- Output gap
- Inflation expectations



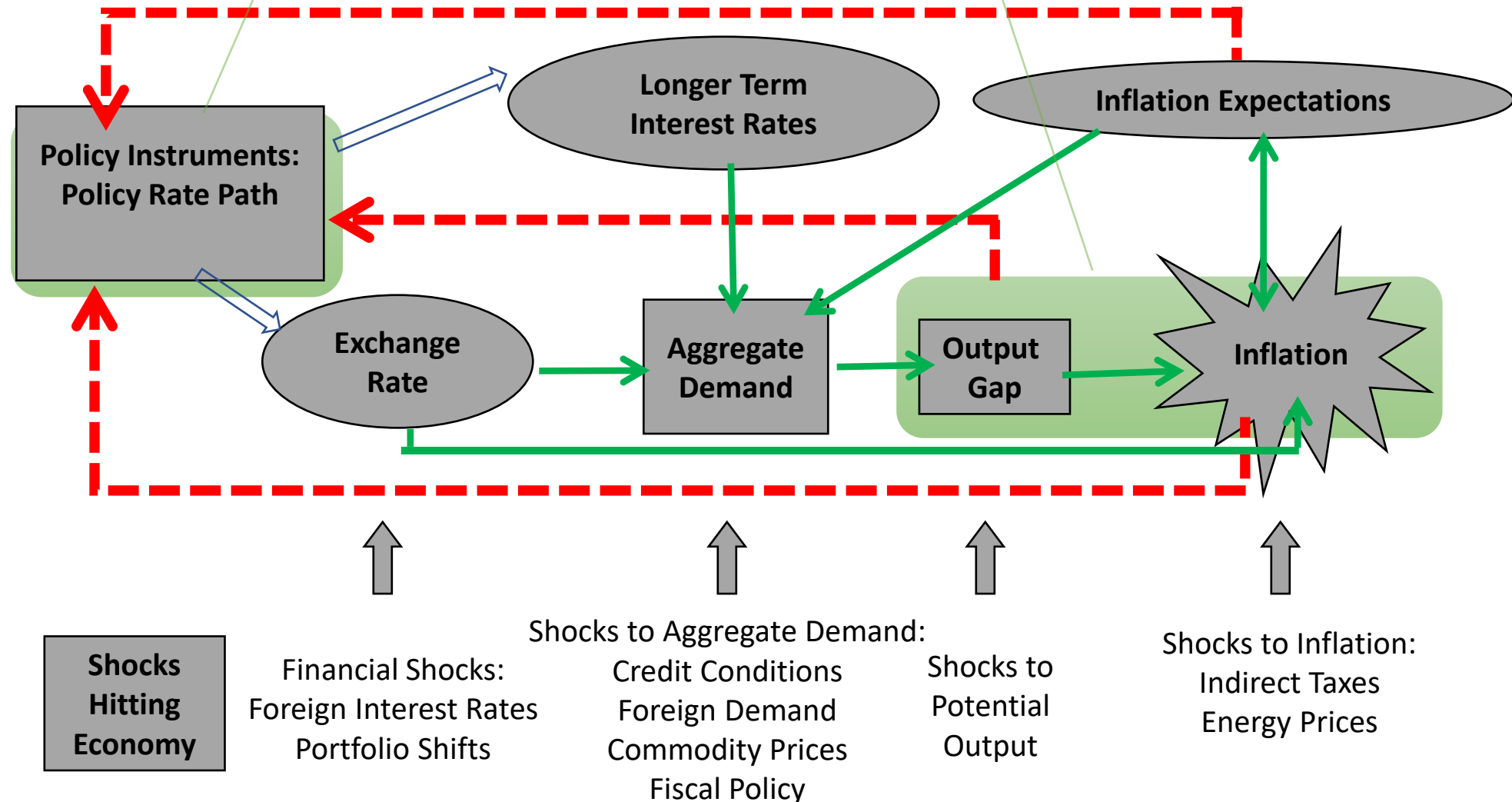
Monetary Transmission Mechanism

Instruments

Unconventional Instruments:

QE, Negative Interest Rates, Funding for Credit, FX Intervention Strategy...

Objectives

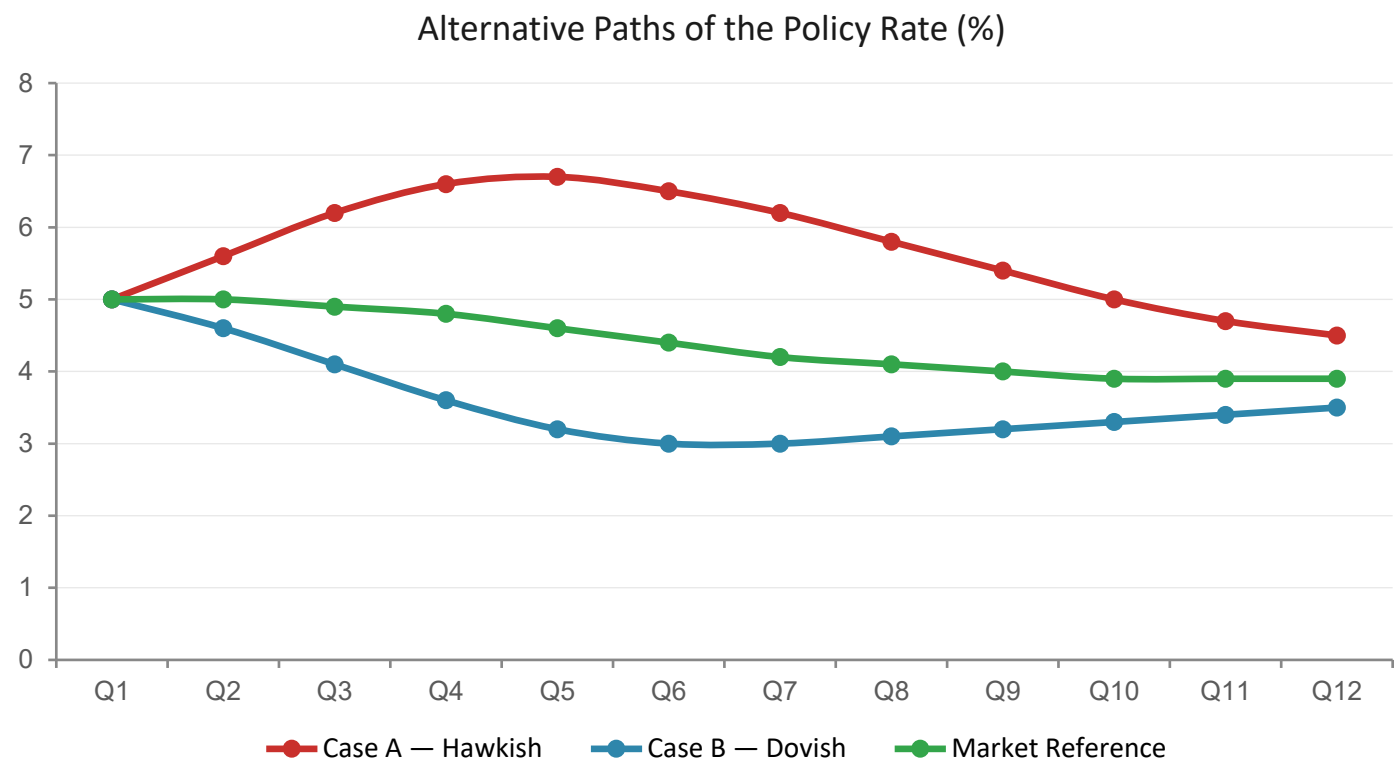




DAY 2 • FRAMEWORK

FPAS Mark II Analytical Structure

Policy is examined across scenarios — no single baseline



CASE A

Hawkish Scenario

Upside inflation risks dominate; tighter policy trajectory keeps expectations anchored.

CASE B

Dovish Scenario

Downside demand risks dominate; easier trajectory supports growth and employment.

MARKET REF.

External Benchmark

Market-implied path used as an external anchor and discipline for the analysis.



DAY 3 • WEDNESDAY, JUNE 10

Semi-Structural Quarterly Projection Models

The workhorse model for central banks — economic structure with policy-relevant behavior



Economic Structure & Intuition

Behavioral equations with clear economic interpretation — not a black-box time series model.



Inflation, Output Gap, Interest Rates

Coherent dynamics linking aggregate demand, the Phillips curve, and monetary policy transmission.



Policy Rules & Reaction Functions

Rule-based policy behavior that supports systematic, transparent decision-making across scenarios.



Risk & Uncertainty in Projections

Scenario-ready design — the model is built to support Case A/B analysis, not a single baseline.



DAY 3 • QPMs

Unobserved Variables & Applied Judgment

Estimating what can't be directly measured

KEY UNOBSERVED VARIABLES

- **Potential output**
The economy's non-inflationary capacity
- **Neutral interest rate (r^*)**
The rate consistent with stable inflation
- **Country risk premium**
Wedge between domestic and foreign yields
- **NAIRU**
Unemployment consistent with stable inflation
- **Inflation expectations**
Forward-looking behavior of price-setters



FRAMEWORK SPOTLIGHT

Sticky/Flexible Price Inflation

- Sticky prices best reflect underlying price-setting behavior of the economy
- Non-tradeable services — e.g., haircut prices — are informative for long-term expectations
- Credibility shows up in the persistence and anchoring of sticky-price inflation
- Central Bank Sticky Price Inflation Indicator (CBSPII) supports credibility measurement



DAY 4 • THURSDAY, JUNE 11

DynareJulia for FPAS Mark II

A modern, open-source platform for macroeconomic modeling

WHAT IS DYNAREJULIA?

A rewrite of Dynare in the Julia programming language — developed as a modern, open-source, free-of-charge platform for macroeconomic modeling.

FOUR OBJECTIVES

Transparent — Open modeling environment

Performant — Improved speed and flexibility

Reproducible — Well-documented workflows

Accessible — No licensing barriers for public institutions

DAY 4 AGENDA

Implementing QPMs in DynareJulia

- Building the core QPM equations in Julia syntax
- Nonlinear dynamics and occasionally binding constraints
- Endogenous credibility and risk considerations
- Solving Mixed Complementarity Problems (MCPs)
- Transparent and reproducible model implementation



DAY 5 • FRIDAY, JUNE 12

Capstone: A Working QPM from Scratch

By the end of the week, each participant has a model ready to produce policy-relevant scenarios

01

Code

Core equations for a macro-consistent projection

02

Calibrate

Model parameters tailored to a country of interest

03

Simulate

Baseline and alternative scenario paths

04

Deliver

Ready to support policy trade-off analysis

Participants leave the week with a working model, the history of best practices, and readiness to begin producing FPAS Mark II scenarios.



Learning Outcomes

By the end of Week 1, participants will be able to:



Apply FPAS Mark II workflows

Understand the institutional logic, roles, and forecasting cycle behind Mark II — and how data, judgment, and scenarios translate into policy-ready analysis.



Trace the history of IT frameworks

Recite the evolution of central bank analytical frameworks supporting inflation targeting, and explain why the folly of baselines led to the Mark II shift.



Operate QPMs in Dynare and DynareJulia

Implement, calibrate, and run semi-structural Quarterly Projection Models in both Dynare/Matlab and the new DynareJulia platform.



Produce scenario-ready analysis

Leave with a working model and the practical skills to begin producing Case A / Case B scenarios for your own institution.



Great Things are Coming.



Website

www.thebetterpolicyproject.org



E-mail

douglaslaxton@thebetterpolicyproject.org



AI-Powered Learning

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Venue

Chulalongkorn University · Bangkok, Thailand
